SETUP & USER GUIDE | VERSION 1.0

Invisible Cloak Pro

How to download, install and run the project from the GitHub ZIP archive

PYTHON · OPENCV · NUMPY

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Repository name used in this guide
Commands below use `invisible-cloak-pro` as the repository name and `invisible_cloak_pro.py` as the main script. If you named them differently on GitHub, substitute your own names.

01 Project Overview

Invisible Cloak Pro is a real-time computer-vision application that creates an "invisibility cloak" effect through a webcam. Anything you wear in the chosen colour (green by default) is replaced, frame by frame, with a previously captured picture of the empty background. The result looks as if the cloth is transparent.

How it works

Stage	What the code does	OpenCV function
1. Capture background	Records the empty scene after a 3-second countdown	`VideoCapture.read()`
2. Colour conversion	Converts each live frame from BGR to HSV colour space	`cv2.cvtColor`
3. Segmentation	Builds a binary mask of every pixel inside the HSV range	`cv2.inRange`
4. Mask clean-up	Closes gaps, removes noise, slightly expands the mask	`morphologyEx, dilate`
5. Replacement	Cloak pixels come from the saved background, other pixels stay live	`bitwise_and, add`

Key features

- Live HSV sliders in a separate control window, so any cloth colour can be tuned on the fly
- Keyboard shortcuts: recapture background, pause, toggle mask preview, toggle the information panel, save screenshot
- Real-time FPS counter and an on-screen information panel with developer details
- Screenshots are saved automatically with timestamps into an `output/` folder

02 System Requirements

> **Install opencv-python, not opencv-python-headless**
> The headless build has no GUI support. Windows, sliders and live preview will fail with an error.

Item	Requirement
Operating system	Windows 10/11, macOS 11 or later, or a desktop Linux distribution (Ubuntu, Fedora, etc.)
Python	Python 3.8 or newer (3.10 - 3.12 recommended)
Libraries	opencv-python and numpy (installed in step 07)
Hardware	A working webcam (built-in or USB) and at least 4 GB RAM
Display	A desktop session. The app opens GUI windows, so it will not run over a plain SSH or headless server
Extras	A green cloth or green chart paper, a plain wall behind you, and a steady camera position

03 What Is Inside the ZIP

After extraction the folder should look like this. The `output/` folder is created automatically the first time the program runs.

```
invisible-cloak-pro/
|-- invisible_cloak_pro.py   <- main application
|-- requirements.txt         <- Python dependencies
|-- README.md                <- project description
|-- .gitignore               <- ignores venv/ and output/
|-- assets/
|     |-- invisible_cloak_pro_output.png
|     \-- invisible_cloak_app_window.png
\-- output/                  <- screenshots appear here (auto-created)
```

04 Downloading from GitHub

Option A: Download as a ZIP (no Git needed)

1. **Open the repository**
   Go to `https://github.com/SUMEETKUMARMONTI/invisible-cloak-pro` in any browser.

2. **Click the green Code button**
   It is located at the top-right of the file list, just above the files.

3. **Choose Download ZIP**
   Your browser saves `invisible-cloak-pro-main.zip` (usually in the Downloads folder).

4. **Wait for the download to finish**
   The archive is small, typically under a few MB.

Option B: Clone with Git (recommended for developers)

```
git clone https://github.com/SUMEETKUMARMONTI/invisible-cloak-pro.git
cd invisible-cloak-pro
```

If you cloned the repository you can skip section 05 and go straight to section 06.

05 Extracting the ZIP File

System	How to extract
Windows	Right-click the ZIP, choose **Extract All...**, pick a folder (for example `C:\Projects`) and click **Extract**.
macOS	Double-click the ZIP. A folder with the same name appears next to it.
Linux	Run `unzip invisible-cloak-pro-main.zip` in a terminal, or right-click and choose **Extract Here**.

> **Tip**
> Extract the files first. Running the script from inside the ZIP preview causes path and permission errors. Avoid folders with unusual characters or very long names.

Now open a terminal inside the extracted folder:

System	How to open the terminal in the folder
Windows	Open the folder in File Explorer, click the address bar, type `cmd` and press Enter. (Or Shift + right-click, **Open in Terminal**.)
macOS	Right-click the folder, **Services**, **New Terminal at Folder**. Or type `cd ` in Terminal and drag the folder into it.
Linux	Right-click empty space in the folder, **Open in Terminal**.

06 Installing Python

Check whether Python is already installed:

```
python --version        # Windows
python3 --version       # macOS / Linux
```

If the command is not found, or the version is below 3.8, download the installer from python.org/downloads.

> **Windows users**
> On the first screen of the installer, tick **Add python.exe to PATH** before clicking Install Now. Without it the `python` and `pip` commands will not work.

07 Virtual Environment & Dependencies

A virtual environment keeps this project's libraries separate from the rest of your system. It is optional but strongly recommended.

Step 1: Create and activate the environment

System	Create	Activate
Windows (cmd)	`python -m venv venv`	`venv\Scripts\activate`
Windows (PowerShell)	`python -m venv venv`	`venv\Scripts\Activate.ps1`
macOS / Linux	`python3 -m venv venv`	`source venv/bin/activate`

When active, your prompt starts with `(venv)`.

> **PowerShell blocked the script?**
> Run `Set-ExecutionPolicy -Scope Process RemoteSigned`, then activate again. This only affects the current window.

Step 2: Install the libraries

```
python -m pip install --upgrade pip
pip install -r requirements.txt
```

If there is no `requirements.txt`, install directly:

```
pip install opencv-python numpy
```

Step 3: Verify the installation

```
python -c "import cv2, numpy; print(cv2.__version__, numpy.__version__)"
```

A line with two version numbers means the setup is complete.

08 Running the Application

```
python invisible_cloak_pro.py        # Windows
python3 invisible_cloak_pro.py       # macOS / Linux
```

What you will see

1. **Console banner**
   The terminal prints the initialisation banner and `[SUCCESS] Webcam initialized.`

2. **Two windows open**
   Invisible Cloak Pro (camera view) and **HSV Color Controls** (six sliders).

3. **3-second countdown**
   The camera view shows *Capturing background in 3...2...1*. Step completely out of the frame during this time.

4. **Effect is live**
   Re-enter the frame wearing the green cloth. The cloth region now shows the saved background.

> **macOS camera permission**
> On first launch, macOS asks whether Terminal (or your editor) may use the camera. Click **OK**. If you clicked Don't Allow, go to System Settings, Privacy & Security, Camera, and enable it for your terminal app.

09 Using the Application

Recommended workflow

- Place the camera on a stable surface or tripod. Any movement after the capture breaks the illusion.
- Light the room evenly and avoid green objects behind you.
- Capture the background, hold the cloth in front of you and move slowly at first.
- Press **M** to check the mask. The cloth should be solid white and everything else black. Adjust the sliders if not.

Keyboard shortcuts

Key	Action
Q or ESC	Exit the application and release the camera
B	Recapture the background (3-second countdown again)
S	Save the current output as a PNG in the output/ folder
M	Toggle between the final result and the black/white mask preview
H	Show or hide the information panel
SPACE	Pause or resume the live feed

Click on the camera window first so that it has keyboard focus, otherwise the keys will not register.

10 Tuning the HSV Colour Range

HSV stands for Hue, Saturation and Value. OpenCV uses hue from 0 to 179, and saturation and value from 0 to 255. The default values detect green. For other colours, use the six sliders and watch the mask preview (key M).

Cloth colour	Lower H, S, V	Upper H, S, V
Green (default)	`50, 80, 50`	`90, 255, 255`
Blue (starting point)	`95, 80, 50`	`130, 255, 255`
Yellow (starting point)	`20, 100, 100`	`35, 255, 255`

Red cloth
Red wraps around both ends of the hue scale (about 0-10 and 170-179), so the single-range mask in this version cannot capture all of it. Use green or blue, or extend the code to combine two masks.

Tuning tips

- **Holes in the cloak:** lower the Lower Saturation or Lower Value slider.
- **Background patches turning invisible:** raise Lower Saturation, or narrow the hue range.
- **Flicker at the edges:** improve the lighting; shadows change the cloth's apparent value.

11 Troubleshooting

Problem	Likely cause and fix
Unable to access webcam	Another app (Zoom, Teams, browser) is using the camera, or permission is denied. Close the other app and check privacy settings. Try changing `CAMERA_INDEX` to 1 in the Config class for an external camera.
'python' is not recognized	Python is not on PATH. Reinstall with **Add to PATH** ticked, or use `py` on Windows.
ModuleNotFoundError: cv2	The libraries are not installed, or the virtual environment is not active. Activate it and run `pip install opencv-python numpy`.
Windows do not open / GUI error	The headless OpenCV build is installed. Run `pip uninstall opencv-python-headless`, then `pip install opencv-python`.
Black or frozen video on Windows	Some webcams are slow to initialise. Wait a few seconds, or try another camera index.
Low FPS	Lower FRAME_WIDTH / FRAME_HEIGHT in Config (for example 640 x 480), close other heavy programs, and improve lighting.
Keys do nothing	The camera window does not have focus. Click on it once.
Ghost outline of the person	The background changed or the camera moved. Press B and recapture while out of frame.
Screenshot not saved	Check that the program has write access to its folder, and that you pressed S after the first frame appeared.

12 Publishing the Project on GitHub

Steps for creating the repository that this guide refers to.

1. **Create the repository**
   On github.com click **New**, name it `invisible-cloak-pro`, choose Public, and add a short description.

2. **Prepare the folder**
   Place the script, `requirements.txt`, `README.md`, `.gitignore` and the `assets/` images in one folder.

3. **Push the code**
   Run the commands below from inside that folder.

4. **Add topics**
   In the repository sidebar add topics: opencv, python, computer-vision, invisible-cloak, hsv.

```
git init
git add .
git commit -m "Initial commit: Invisible Cloak Pro"
git branch -M main
git remote add origin https://github.com/SUMEETKUMARMONTI/invisible-cloak-pro.git
git push -u origin main
```

Contents of requirements.txt

```
opencv-python>=4.5
numpy>=1.21
```

Contents of .gitignore

```
venv/
output/
__pycache__/
*.pyc
```

No Git installed?
On the repository page choose **Add file**, then **Upload files**, drag the files in and click **Commit changes**.

A Appendix: Expected Output

The diagram below illustrates the three stages of the pipeline: the original frame, the HSV mask of the cloth, and the final frame in which the cloth region has been replaced by the stored background.

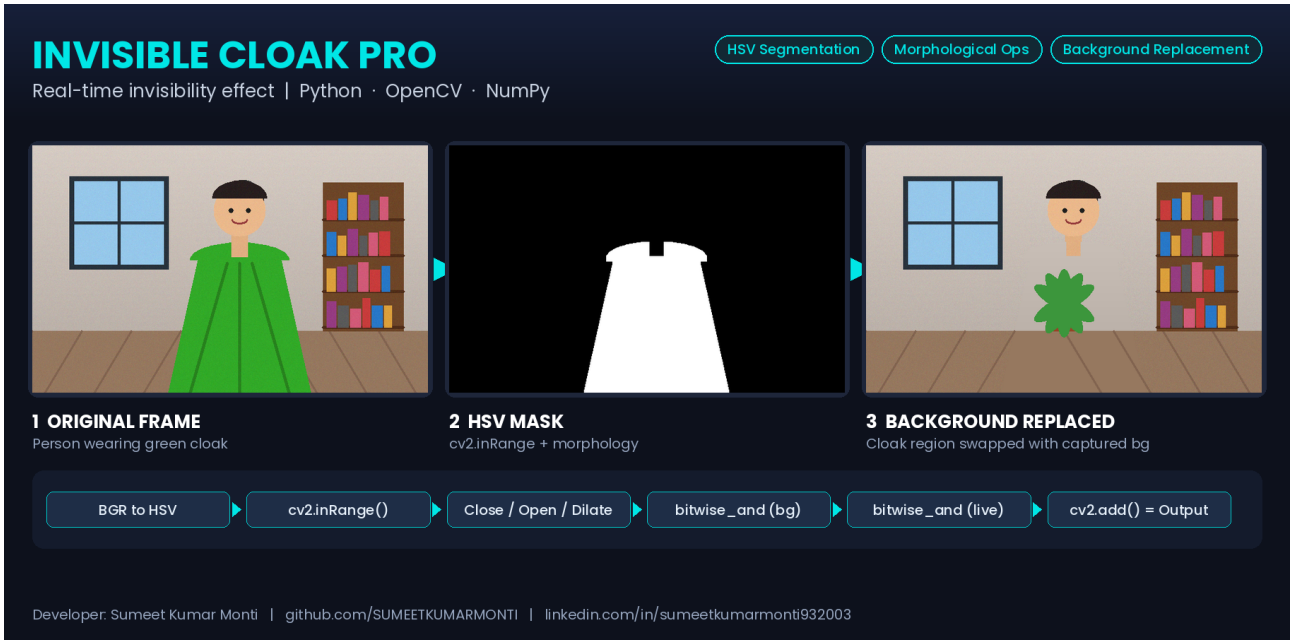


Figure 1. Pipeline illustration generated by running the project's processing steps on a synthetic scene.



Figure 2. Application window with the information panel (illustrative).

> **Your real result**
> Press **S** while the program runs to save real screenshots to the output/ folder, and use them in your README in place of these illustrations.